School of Natural and Computing Sciences

Master's Dissertation

Smart Home Sensing, Time-Series Analytics, and a Digital Twin for Indoor Climate & Energy

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Abstract

The increasing adoption of smart home technologies has enabled continuous monitoring of indoor environmental conditions, including temperature, humidity, light intensity, and communication signals. However, transforming raw sensor data into meaningful insights remains a challenge, particularly when dealing with high-frequency time-series data and system reliability. This project aims to develop an integrated framework for smart home sensing by combining real-time data acquisition, time-series analytics, and a digital twin representation of indoor climate and energy-related conditions.

Data was generated by an ESP32 microcontroller-based smart farmhouse sensing system and transmitted using the Message Queuing Telemetry Transport (MQTT) protocol. The proposed framework acquires, stores, visualises, and analyses streaming IoT telemetry to enable real-time monitoring and predictive analytics. A live dashboard and mobile monitoring interface were implemented to display current sensor values, while historical data was analysed to identify temporal patterns, system behaviour, and reliability issues. Time-series techniques were applied to investigate seasonality, relationships between variables, and delayed effects. In addition, anomaly detection and rule-based monitoring approaches were used to assess sensor performance and communication stability. The results demonstrate clear daily patterns in light intensity and temperature, as well as measurable relationships between environmental variables. Lagged cross-correlation analysis showed that changes in light intensity were associated with later temperature variations, which is consistent with delayed thermal response. Some sensor channels remained constant or unreliable, highlighting practical limitations in sensor behaviour. Monitoring methods also identified temporary communication degradation and candidate sensor anomalies.

Overall, this study shows that combining real-time data streaming with time-series analytics enables effective monitoring and interpretation of indoor environmental conditions. The developed framework provides a foundation for a digital-twin-inspired monitoring system of the smart home environment, supporting system understanding, sensor-health assessment, and future predictive monitoring.

Introduction

Smart home technologies have become increasingly common in recent years, especially with the growth of low-cost sensors and IoT platforms (Atzori et al., 2010). Devices are now able to continuously measure indoor environmental conditions such as temperature, humidity, and light, generating large amounts of time-dependent data. While this creates opportunities for better monitoring and control of indoor climate and energy usage, it also introduces challenges in terms of data interpretation and system reliability.

In many cases, sensor data is collected and displayed, but not analysed in a meaningful way. Simply visualising values is not enough to understand how the system behaves over time, how variables interact with each other, or whether the sensors themselves are functioning correctly. Indoor environments are dynamic, and the relationships between variables are often not immediate. For example, changes in light intensity can influence temperature with a delay, and humidity levels may respond differently depending on multiple factors. This makes time-series analysis a necessary approach for extracting useful insights from the data (Hyndman and Athanasopoulos, 2021).

Another important aspect is the reliability of the sensing system. In real-world deployments, sensors may produce constant values, experience communication drops, or behave unexpectedly. Identifying these issues is essential, especially if the data is used for decision-making or automation. This is where predictive monitoring and anomaly detection become relevant, as they allow early identification of potential problems in the system (Chandola et al., 2009).

At the same time, the concept of a digital twin provides a useful framework for representing physical environments in a virtual form. By combining real-time data with analytical models, it becomes possible to create a dynamic representation of the indoor environment that reflects its current state and behaviour over time (Tao et al., 2018).

The aim of this project is to develop a smart home sensing system that integrates real-time data collection, time-series analytics, and predictive monitoring into a single workflow. The project focuses on analysing indoor climate data, identifying patterns and relationships between variables, and detecting anomalies related to sensor performance. In doing so, it contributes towards building a practical digital twin for indoor climate and energy monitoring.

This work contributes an integrated IoT–Digital Twin framework that combines live sensing, cloud-based data ingestion, statistical time-series analysis, sensor-health monitoring, and short-term forecasting within a unified architecture. Unlike systems that focus solely on data acquisition or visualisation, the proposed framework integrates the complete workflow from physical sensing and real-time monitoring to historical analytics and predictive assessment.

Data

The data used in this project was generated by a smart home sensing system based on an ESP32 microcontroller. The device was configured to measure multiple indoor environmental parameters, including temperature, relative humidity, light intensity, soil moisture, water level, rainfall indicator, motion detection, button state, ultrasonic distance, and signal strength (RSSI). These measurements were generated at approximately one-second intervals, producing a continuous stream of time-stamped observations. Sensor data was transmitted in real time using the MQTT protocol to a cloud-based broker (HiveMQ). Each message was structured in JSON format and contained both sensor readings and metadata, including sequence number and communication signal strength. The received telemetry was stored for later time-series analysis. This ensured that all incoming telemetry was persistently recorded for further analysis.

The resulting dataset consists of time-series data indexed by UTC timestamps, with each row representing a single observation of all available sensor variables. The data was later aggregated into multiple temporal resolutions, including one-minute and hourly averages, to support different types of analysis. This resampling step allowed both short-term fluctuations and longer-term trends to be examined. Variables such as temperature, humidity, light intensity, and RSSI exhibited meaningful variation and formed the basis for the main analysis.

Overall, the dataset provides a realistic representation of indoor environmental monitoring in a smart home setting, including both dynamic sensor behaviour and practical data quality challenges.

4. Methods

4.1 Physical Sensing System and Smart Farmhouse Setup

The physical sensing system was based on an ESP32 PLUS development board installed within a smart farmhouse model. The ESP32 acted as the central microcontroller, collecting readings from multiple connected sensor modules and transmitting the resulting telemetry data through Wi-Fi. The connected modules included a DHT11 temperature and humidity sensor, a photoresistor for light-intensity measurement, a PIR motion sensor for event detection, an ultrasonic distance sensor, a button module, and additional environmental sensing modules such as soil humidity and water-level sensors. The ESP32 converted these sensor signals into structured telemetry messages, which were then transmitted using the Message Queuing Telemetry Transport (MQTT) protocol to the HiveMQ cloud broker. This physical setup provided the data-generating layer of the project, while the later stages of the pipeline focused on data ingestion, storage, visualisation, and time-series analysis.

Figure 1. ESP32 Smart farmhouse physical setup

Component / signal	Measured quantity or role	Data type	Unit / coding	Use in analysis	Observed behaviour
ESP32 PLUS development board	Central microcontroller for sensor reading and Wi-Fi transmission	Controller	N/A	Data generation and MQTT transmission	Operational platform
DHT11 temperature and humidity sensor	Temperature	Continuous	°C	Time-series analysis, decomposition, forecasting	Varied
DHT11 temperature and humidity sensor	Relative humidity	Continuous	%	Time-series analysis, correlation, forecasting	Varied
Photoresistor	Light intensity	Continuous / analogue	ADC value	Seasonality and lagged correlation	Varied
PIR motion sensor	Motion detection	Binary event	0/1	Event-based analysis	Sporadic activity
Button module	Button/input state	Binary event	0/1	Event-state and reliability analysis	Mostly active / abnormal behaviour
Ultrasonic sensor	Distance	Continuous	cm	Sensor-health monitoring	Mostly stable with occasional outliers
Soil humidity sensor	Soil moisture channel	Continuous / analogue	ADC value	Data-quality check	Mostly constant
Water-level sensor	Water-level channel	Continuous / analogue	ADC value	Data-quality check	Mostly constant
RSSI	Wi-Fi signal strength	Continuous	dBm	Communication-health monitoring	Varied, with signal drops

Table 1. Sensors and components used in the ESP32-based smart farmhouse platform

The table separates continuous environmental variables, binary event-based signals, and communication-health variables used in the time-series and sensor-reliability analysis.

4.2 System Architecture and Data Pipeline

The system was designed as a real-time data pipeline connecting physical sensing, cloud storage, visualisation, alerting, mobile monitoring, and offline data-science analysis. Environmental and event-based data were generated by an ESP32-based smart sensing device and transmitted using the MQTT protocol to a HiveMQ cloud broker, which provides lightweight publish/subscribe communication suitable for Internet of Things applications (Banks and Gupta, 2014).
To support persistent storage, a Python ingestion service was deployed on an Amazon EC2 instance running Ubuntu. The service subscribed to the MQTT telemetry stream, decoded incoming JSON messages, normalised the telemetry fields, and inserted the processed records into a PostgreSQL database. This database provided structured storage for historical time-series data and enabled later export for offline analysis.
For real-time monitoring, PostgreSQL was connected to Grafana, where dashboards were developed to visualise temperature, relative humidity, light intensity, motion status, and RSSI. Grafana alerting rules were also configured to send threshold-based notifications through Gmail and Telegram when abnormal conditions were detected.
Historical telemetry exported from PostgreSQL was then analysed using Python workflows, including time-series analysis, anomaly detection, rule-based sensor health monitoring, and forecasting.
Overall, the architecture combines IoT sensing, MQTT communication, cloud ingestion, relational database storage, Grafana-based visualisation and alerting, mobile monitoring, and Python-based analysis. These components form an initial digital-twin-style monitoring framework for smart home and sensor-health applications.

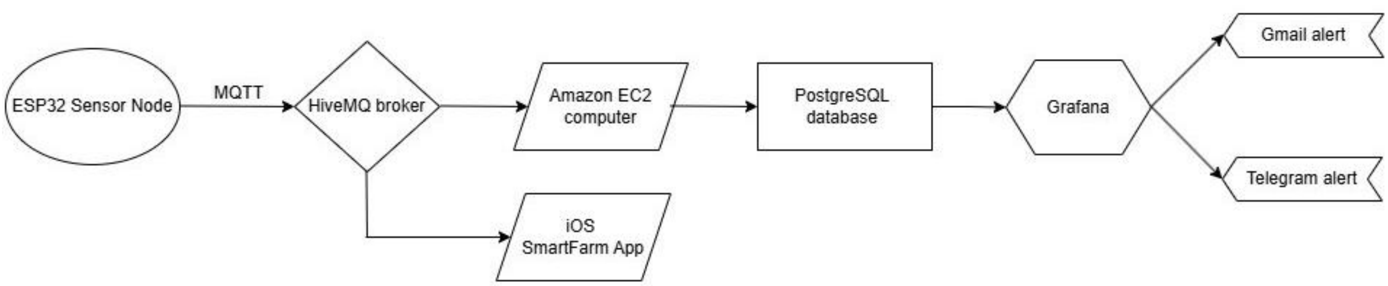

Figure 2. System architecture and data flow of the ESP32-based smart farmhouse monitoring system

Figure 2 shows that sensor telemetry is transmitted from the ESP32 sensor node to the HiveMQ broker using MQTT. The data are then ingested through an Amazon EC2 instance, stored in PostgreSQL, visualised in Grafana, and used for alerting. A mobile application also connects to the MQTT broker for real-time monitoring.
The real-time monitoring layer was implemented using Grafana, which provided an operational view of current sensor values and recent time-series behaviour.

Figure 3. Grafana dashboard used for real-time monitoring of smart farmhouse telemetry

The dashboard in Figure 3 displays current values and recent trends for distance, temperature, humidity, light intensity, Wi-Fi signal strength, and PIR motion status. This interface supports live monitoring of both environmental behaviour and communication health.

A mobile application was developed in Xcode as an additional monitoring interface. The iOS application connected directly to the HiveMQ broker through a WebSocket MQTT interface, allowing real-time display of selected sensor values.

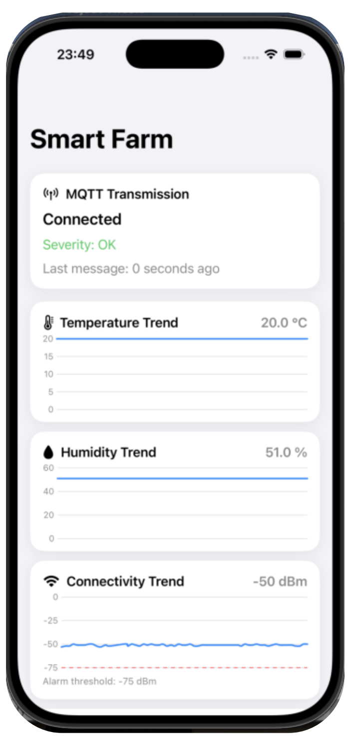

Figure 4. iOS Smart Farm mobile application used for real-time telemetry monitoring

Figure 4 shows the iOS application connected to the HiveMQ broker using MQTT over WebSocket and displayed live telemetry values and trends, including temperature, humidity, connectivity status, MQTT transmission status, and an RSSI alarm threshold.

4.3 Mathematical formulation of the telemetry data

The telemetry stream was treated as a multivariate time series. At each timestamp t, the sensor state was represented as:

$$x_t = (T_t, H_t, L_t, C_t, D_t, P_t, B_t)$$

where T_t is temperature, H_t is relative humidity, L_t is light intensity, C_t is connectivity signal strength measured using RSSI, D_t is ultrasonic distance, P_t is PIR motion state, and B_t is button state. Continuous variables such as temperature, humidity, light intensity, RSSI, and distance were analysed using resampling, descriptive statistics, correlation analysis, seasonal decomposition, anomaly detection, and forecasting. Binary variables such as PIR motion and button state were treated separately because they represent event states rather than continuous physical measurements.

4.4 Data Preprocessing

Before analysis, the dataset was prepared to ensure consistency and usability. All timestamps were converted to a standard UTC datetime format and used as the index of the dataset. The data was then sorted chronologically to preserve temporal order. Basic data quality checks were performed, including the identification of missing values, duplicate records, time gaps, and irregularities in sequence numbers. Missing-value analysis was used to assess whether received telemetry records contained incomplete fields, while time-gap and sequence-number analysis were used to identify possible delayed transmission, packet loss, or device resets. It is important to distinguish between missing values inside received records and missing observations in the telemetry stream. A zero missing-value rate indicates that the received records contained no empty fields, but it does not necessarily prove that no messages were delayed or lost during transmission.

4.5 Resampling Strategy

Because the raw telemetry was recorded at approximately one-second intervals, aggregation was required to analyse longer-term temporal patterns without excessive short-term noise. For continuous variables, resampling was performed by calculating the arithmetic mean within each time window. For a variable x_t and a time window W_j, the resampled value was defined as:

$$\overline{x_j} = \frac{1}{|W_j|} \sum_{t_i \in W_j} x_{t_i}$$

where W_j is the j-th time window and $|W_j|$ is the number of observations inside that window. One-minute averages were used for detailed time-series and seasonal analysis, while hourly averages were used for lagged correlation, rolling monitoring, and forecasting. This reduced short-term noise and allowed longer-term temporal structure to be examined.

The firmware was configured to read and publish sensor telemetry at approximately one-second intervals. Under continuous operation, the expected number of samples per hour was therefore approximately:

$$N_{\text{hour}} = \frac{3600}{1} = 3600\ \text{samples/hour}$$

4.6 Native-Resolution Event-Based Analysis

Binary variables such as PIR motion and button state were analysed separately from continuous variables because they represent event states rather than physical magnitudes. Hourly aggregation can be misleading because a binary signal may remain active for many samples without representing repeated physical events.
For a binary signal b_t in window W_j, the active-state fraction was defined as:

$$\widehat{p_j} = \frac{1}{|W_j|} \sum_{t_i \in W_j} b_{t_i}$$

To detect discrete event starts, edge detection was applied at the native sampling resolution. The PIR channel was active-high, so motion events were identified as upward transitions from 0 to 1. The button channel was active-low, so button press events were identified as downward transitions from 1 to 0:
PIR event at $t_i \Leftrightarrow P_{t_{i-1}} = 0$ and $P_{t_i} = 1$
Button press at $t_i \Leftrightarrow B_{t_{i-1}} = 1$ and $B_{t_i} = 0$
This approach allowed event-based signals to be interpreted as state transitions rather than simple hourly counts.

4.7 Descriptive Statistics and Seasonal Decomposition

Descriptive statistics were computed to summarise the behaviour of the continuous environmental and communication variables, including the mean, standard deviation, minimum, maximum, and quartiles. This step helped identify which sensor channels showed meaningful variation and which remained constant or inactive. This was important because variables such as temperature, humidity, light intensity, and RSSI were suitable for time-series analysis, while constant channels such as soil, water, or rain-related measurements were mainly treated as data-quality findings. To analyse temporal behaviour, seasonal decomposition was applied to selected variables using an additive model:

$$y_t = T_t + S_t + R_t$$

where y_t is the observed time series, T_t is the trend component, S_t is the seasonal component, and R_t is the residual component. Seasonal decomposition was selected because indoor environmental data can contain repeated daily patterns, especially in variables affected by daylight and thermal response. For one-minute data, a daily seasonal period of 1440 was used, corresponding to 1440 minutes per day. This allowed regular daily behaviour to be separated from longer-term trends and irregular residual variation. In this project, the method helped identify normal environmental behaviour and provided residuals for later anomaly detection.

4.8 Correlation and Lagged Correlation Analysis

Relationships between environmental variables were analysed using Spearman correlation. This method was selected because sensor relationships may be monotonic but not strictly linear, and Spearman correlation uses ranked values rather than assuming a linear relationship. It was defined as:

$$\rho_s(X, Y) = corr\big(rank(X), rank(Y)\big)$$

The correlation matrix was used as an exploratory tool to identify associations between variables such as temperature, humidity, light intensity, and RSSI. This helped assess which variables tended to vary together. However, correlation was not interpreted as evidence of causation, especially because time-series variables may share daily cycles and autocorrelation.

To investigate delayed relationships, lagged correlation analysis was performed. This was important because environmental responses are not always immediate. For example, changes in light intensity may occur before changes in temperature due to thermal inertia. Before lagged correlation analysis, variables were normalised using z-score transformation:

$$z_t = \frac{x_t - \bar{x}}{s_x}$$

where $\bar{x}$ is the sample mean and s_x is the sample standard deviation.
Lagged correlation was then computed between two standardised series across a range of time lags *k*:

$$\rho_{xy}(k) = corr\big(z_t^x, z_{t+k}^y\big)$$

In this study, a positive lags indicate that earlier values of the first variable are associated with later values of the second variable. For example, in the light-temperature analysis, a positive lag means that changes in light intensity occurred before changes in temperature. Therefore, lagged correlation helped identify delayed environmental responses, while still being interpreted as temporal association rather than direct causation.

4.9 Residual-Based Anomaly Detection

Anomaly detection was performed using the residuals obtained from seasonal decomposition. This method was selected because raw sensor signals contain normal daily patterns, especially for variables such as light intensity and RSSI. Detecting anomalies directly from the raw signal could incorrectly treat normal daily variation as abnormal. Therefore, anomalies were identified from the residual component, which represents the part of the signal not explained by the estimated trend or regular daily seasonal pattern. The residual was calculated as:

$$R_t = y_t - T_t - S_t$$

The residual represents the part of the signal that is not explained by the estimated trend or regular daily seasonal pattern. Therefore, sudden drops, spikes, or irregular deviations appear as large positive or negative residual values.
Because the residual component is approximately mean-centred, anomalies were identified using a three-standard-deviation threshold applied to the residual values, which is equivalent to a residual z-score threshold of ±3.

For each residual value R_t, a residual z-score was calculated as:

$$z_t^{(R)} = \frac{R_t - \mu_R}{\sigma_R}$$

where μ_R is the mean of the residuals and σ_R is the residual standard deviation. This z-score measures how many standard deviations each residual is away from the typical residual behaviour.
An observation was flagged as a candidate anomaly if:

$$\left|z_t^{(R)}\right| > 3$$

This means that a point was treated as unusual if its residual was more than three standard deviations away from the mean residual. In practical terms, the method detected sudden deviations from the expected pattern rather than simply detecting high or low raw values. For RSSI, large negative residuals may indicate sudden signal degradation or temporary communication instability. For light intensity, large residuals may indicate sudden environmental changes or irregular sensor behaviour. The three-standard-deviation threshold was used as a simple and interpretable statistical rule, especially because labelled fault data were not available. However, detected points were interpreted as candidate anomalies rather than confirmed failures, because time-series residuals may be autocorrelated and non-Gaussian. Therefore, residual-based anomaly detection was used as an early-warning and diagnostic method rather than definitive proof of sensor failure.

4.10 Rule-Based Sensor Health Monitoring

Rule-based sensor health monitoring was implemented to detect weak communication, instability, flat-line behaviour, and saturation in the main monitored variables. The analysis used hourly resampled data with a 24-hour rolling window. For communication health, RSSI was monitored using a 24-hour rolling mean. A weak-signal alert was defined when the rolling mean fell below −75 dBm, allowing sustained communication degradation to be distinguished from short-term fluctuations. For temperature and humidity, rolling variance was used to identify abnormal instability or possible flat-line behaviour. High variance was flagged using an upper threshold defined as the mean rolling variance plus three standard deviations. Very low variance was treated as possible flat-line or low-activity behaviour. For the light sensor, flat-line behaviour was assessed using the rolling mean of absolute differences between consecutive hourly readings.

5. Results

5.1 Data Quality and Overview

The dataset showed high completeness, with no missing values detected across all variables. Time intervals between observations were consistent, indicating stable data transmission. However, sequence number analysis revealed occasional irregularities, suggesting minor packet loss or device resets.
Several environmental variables demonstrated meaningful variation, including temperature, humidity, light intensity, and RSSI. In contrast, soil moisture, water level, and rainfall remained constant throughout the observation period, indicating inactive sensors or absence of external stimuli.

5.2 Time-Series Behaviour and Seasonality

Clear daily patterns were observed in the environmental data. Light intensity exhibited the strongest periodic behaviour, with values rising during daytime and dropping to near zero at night. Figures 5–7 show the seasonal decomposition of light intensity at different time scales. The full-period decomposition highlights the repeated daily structure across the dataset, while the three-day and one-day views provide closer inspection of local trend and residual variation. The seasonal component remains clearly visible, confirming that light intensity was the variable with the strongest daily periodic structure. Temperature followed a similar but smoother temporal pattern, suggesting delayed thermal response to light conditions. Humidity showed moderate variability with weaker seasonal structure. Additional seasonal decompositions for other variables are provided in the Supplementary Material. (Figures S3–S5).

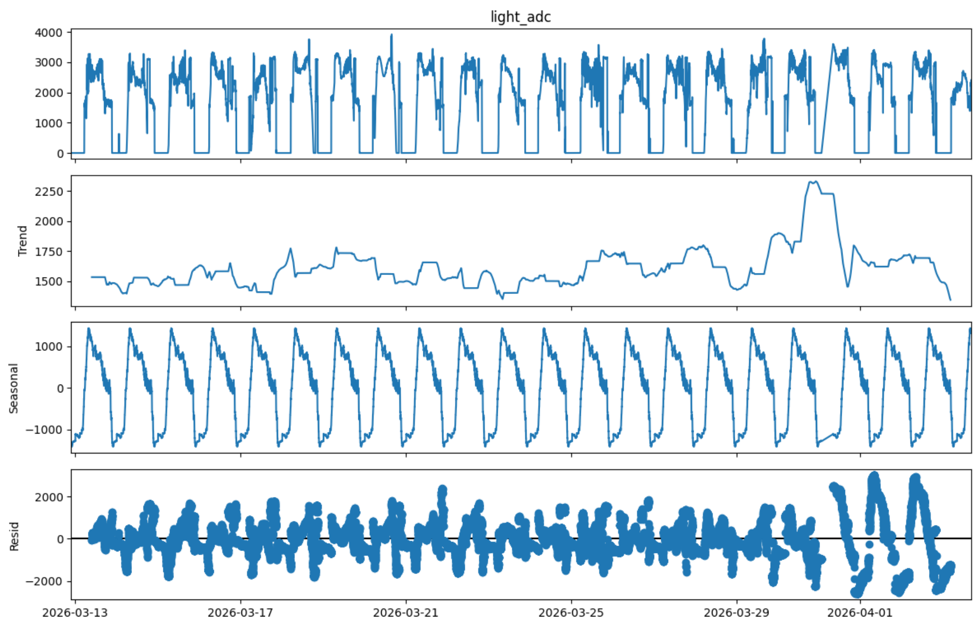

Figure 5. Seasonal decomposition of light intensity over the full observation period

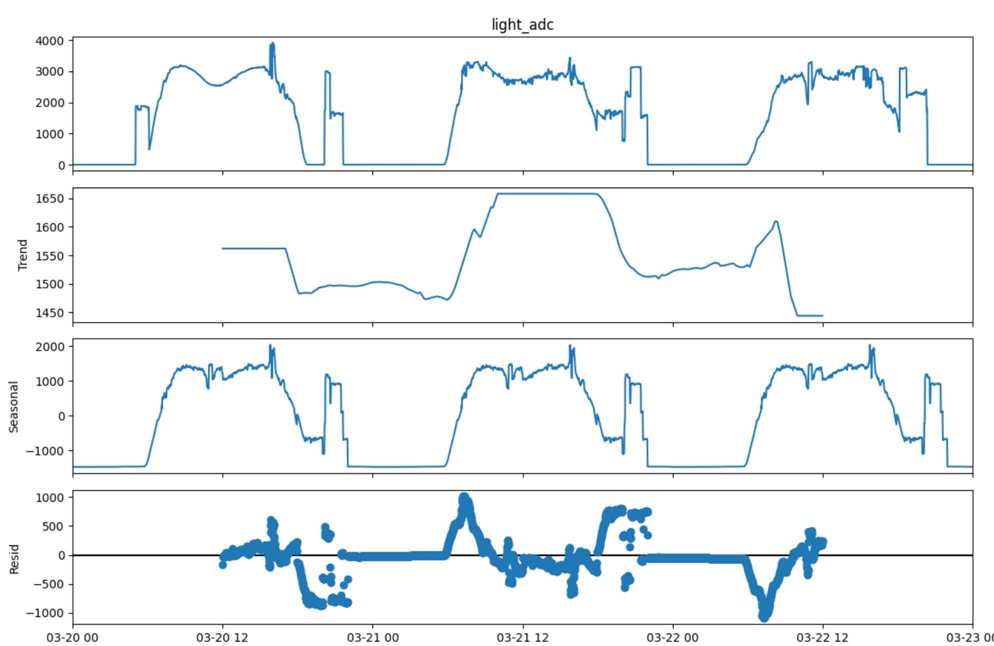

Figure 6. Three-day seasonal decomposition of light intensity

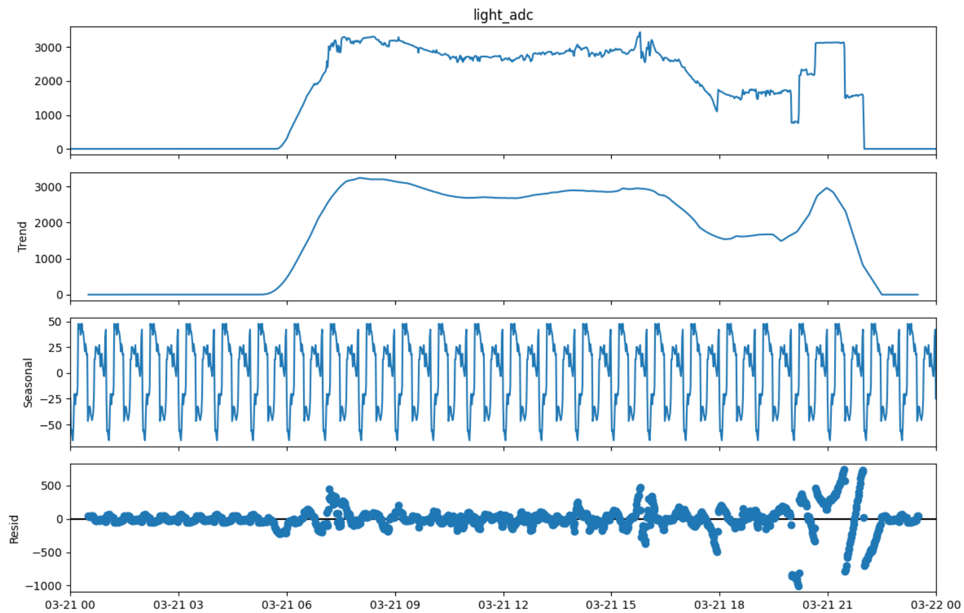

Figure 7. One-day seasonal decomposition of light intensity

5.3 Correlation Analysis

The relationships between variables were examined using Spearman correlation.
As shown in **Figure 8**, temperature and humidity exhibit a moderate positive correlation (~0.58), indicating that they vary together. Light intensity shows a weaker relationship with humidity but contributes to temperature variation.
Variables with constant values were excluded from interpretation, as they do not contribute meaningful information.

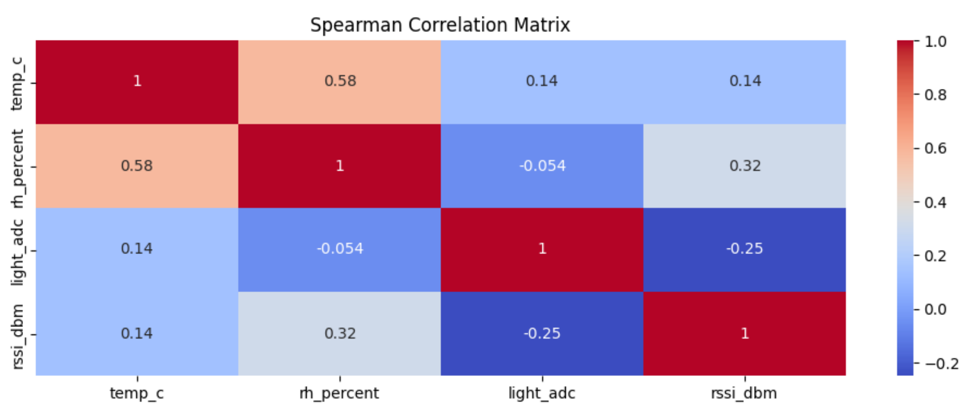

Figure 8. Spearman correlation matrix illustrating relationships between environmental variables. Temperature and humidity exhibit the strongest correlation.

5.4 Lagged Relationships

Lagged cross-correlation analysis was used to examine whether changes in one environmental variable were associated with delayed changes in another variable. **Figure 9** shows the correlation between hourly z-score-normalised variables across lags from 0 to 24 hours.
For the light-temperature pair, the highest positive correlation occurred at approximately 5–6 hours. This means that temperature values were most strongly associated with light-intensity values observed around 5–6 hours earlier. This result is physically consistent with thermal inertia, where light exposure may affect indoor temperature after a delay.
Temperature and humidity showed the strongest association at zero lag, suggesting more synchronous behaviour. The relationship between light and humidity remained weak across most lags. These results were interpreted as lagged temporal associations rather than evidence of direct causation. This caution is necessary because environmental time series may contain autocorrelation and shared daily cycles, which can influence lagged correlation patterns.

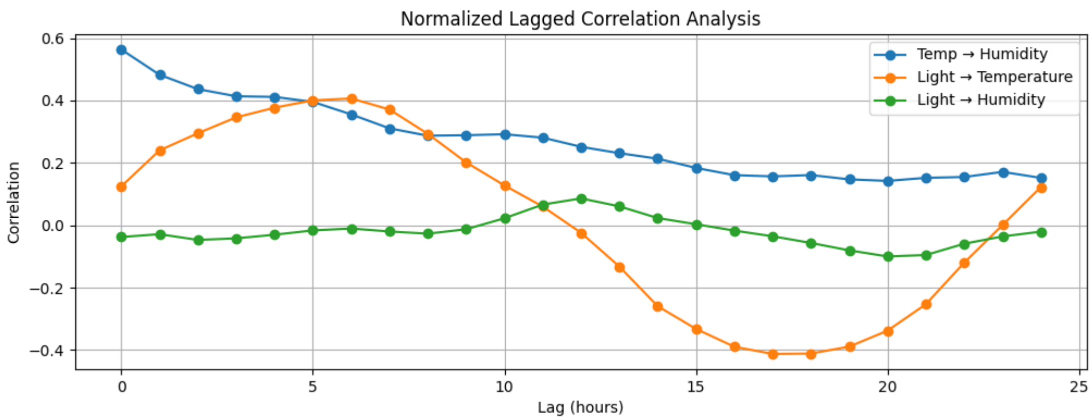

Figure 9. Lagged Cross-Correlation Analysis of Standardised Variables. Peak correlation between light and temperature occurs at a lag of approximately 5–6 hours.

5.5 Anomaly Detection

Anomaly detection identified several irregular events in the dataset. In **Figure 10,** RSSI anomalies are visible as sharp drops in signal strength, indicating temporary communication degradation. These events may be caused by interference or connectivity instability.
Light sensor anomalies were also detected but were less consistent and therefore not included in the main analysis.

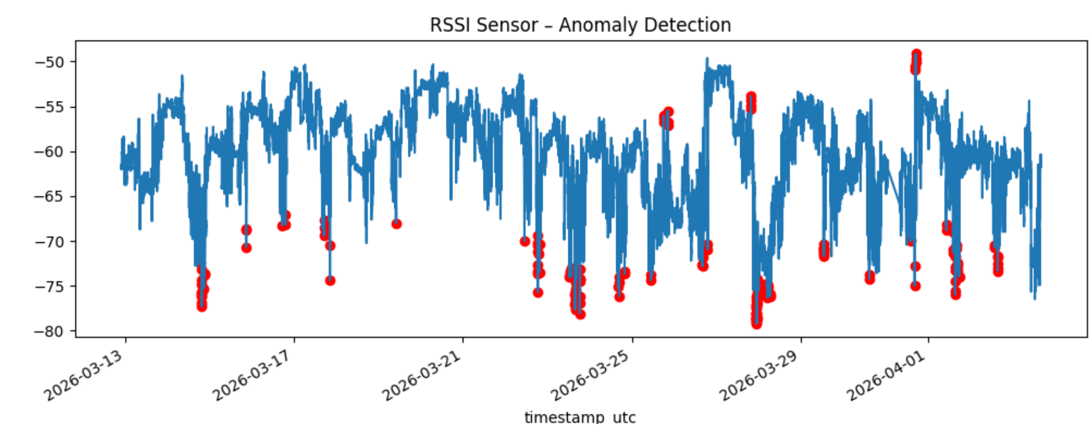

Figure 10. RSSI anomaly detection showing periods of significant signal degradation, highlighted as deviations from normal behaviour

5.6 Event-Based Behaviour

Event-based signals were analysed at their native binary resolution rather than using hourly aggregation. This was necessary because binary signals can remain active for several samples, which may create high hourly counts without representing repeated physical events. The PIR motion channel was treated as active-high, where 0 indicates no motion and 1 indicates motion detected. The button channel was treated as active-low, where 1 indicates not pressed and 0 indicates pressed. Figure 11 shows two button press events on the selected day, identified as downward transitions from 1 to 0.

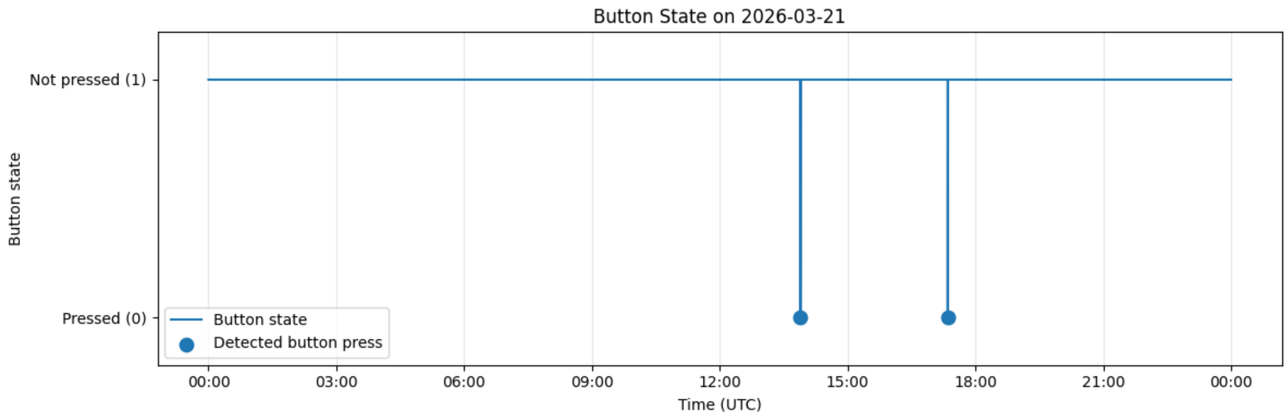

Figure 11. Native-resolution button state on the selected event day

Figure 12 shows PIR motion events, identified as upward transitions from 0 to 1. These results confirm that native-resolution edge detection provides a clearer interpretation of event-based telemetry than hourly counts.

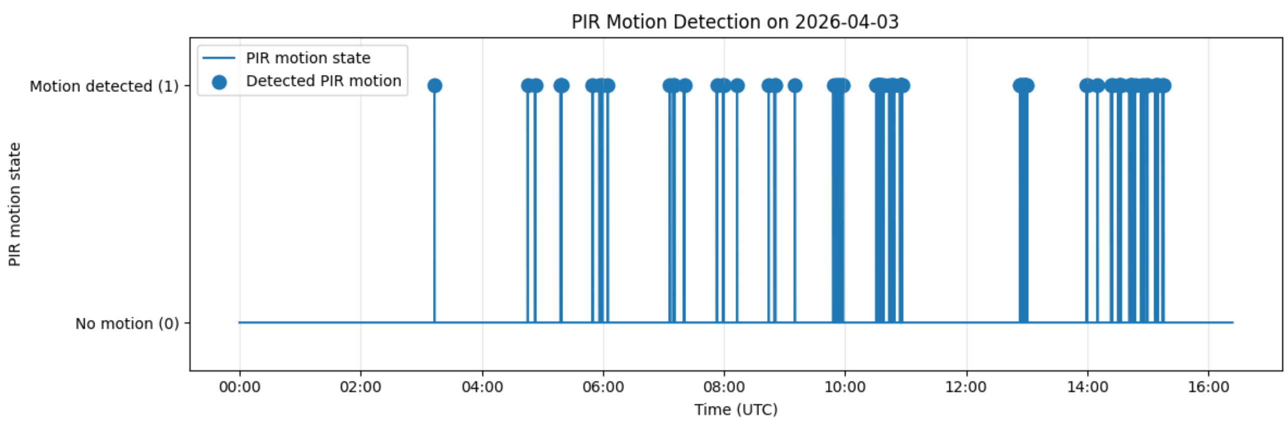

Figure 12. Native-resolution PIR motion signal on the selected event day

5.7 Rule-Based Sensor Health Monitoring

Rule-based sensor health monitoring was used to assess communication quality and sensor stability. RSSI monitoring showed short-term signal drops, but the 24-hour rolling mean generally remained above the weak-signal threshold of −75 dBm, indicating that sustained communication degradation was limited. Rolling-variance analysis of temperature and humidity identified occasional periods of increased variability, which may reflect environmental fluctuation or sensor noise. Light-sensor

monitoring also showed periods of low variation and possible flat-line behaviour, supporting the need to check sensor reliability before interpretation.
Overall, the rule-based results indicate that the system operated mostly reliably, but intermittent communication and sensor irregularities highlight the importance of continuous monitoring. Additional predictive-monitoring figures are provided in the Supplementary Material. (Figures S6–S10).

5.8 Forecasting Performance

Forecasting was evaluated using a decomposition-based ARIMA approach and compared with naive and seasonal-naive baselines. The final 24 hours of data were used as a hold-out test period. Table 2 summarises the forecasting performance for temperature, humidity, and RSSI.

Target	Model	MAE	RMSE	R^2
Temperature (°C)	Naive persistence	1.476	1.604	-5.564
Temperature (°C)	Seasonal naive	0.727	1.169	-2.487
Temperature (°C)	Decomposition + ARIMA residual	1.093	1.308	-3.366
Humidity (%)	Naive persistence	6.495	6.751	-12.437
Humidity (%)	Seasonal naive	2.934	3.703	-3.043
Humidity (%)	Decomposition + ARIMA residual	2.596	3.377	-2.363
RSSI (dBm)	Naive persistence	4.095	4.495	-0.248
RSSI (dBm)	Seasonal naive	3.378	4.742	-0.388
RSSI (dBm)	Decomposition + ARIMA residual	2.791	3.533	0.229

Table 2. Forecasting performance for naive, seasonal-naive, and decomposition-based ARIMA models over the final 24-hour test period

Table 2 shows that model performance varied by variable. For temperature, the seasonal-naive baseline achieved the lowest error, indicating that repeating the previous daily pattern was more effective than modelling the residual component with ARIMA. For humidity and RSSI, the decomposition-based ARIMA model achieved the lowest MAE and RMSE values.
Overall, decomposition-based residual forecasting improved short-term prediction for some variables, particularly humidity and RSSI, but it was not consistently superior across all targets. In this project, forecasting supports the digital-twin concept by extending the system from real-time monitoring towards near-future estimation of environmental and communication conditions. However, the negative R^2 values for temperature and humidity indicate limited predictive performance over the short test period, so the results should be interpreted as exploratory.

6. Discussion

6.1 Interpretation of Environmental Dynamics

The strong daily periodicity observed in light intensity confirms that the sensing system is capturing realistic environmental behaviour. This periodic pattern propagates into temperature, which responds with a delay due to thermal inertia.
The lagged correlation analysis provides further insight into this relationship. The observed delay of approximately 5–6 hours between light intensity and temperature suggests that heat accumulation within the indoor environment is gradual rather than immediate. This is consistent with physical expectations, as building materials absorb and release heat over time. In contrast, humidity appears to

respond more directly to temperature changes, with little to no lag, indicating a tightly coupled relationship between these variables.

6.2 Reliability of Sensor Data

While the overall data quality was high, several inconsistencies were identified. The presence of constant values in soil moisture, water level, and rainfall sensors suggests that these sensors were either inactive or not exposed to changing conditions. As a result, they did not contribute meaningful information to the main analysis. Native-resolution edge detection therefore provided a more reliable interpretation of button and PIR motion events.

Similarly, RSSI anomalies indicated periods of degraded communication quality. Although these events were relatively limited, they highlight the importance of monitoring not only environmental conditions but also system connectivity. Overall, these observations show that raw sensor data cannot be assumed to be fully reliable and must be validated before interpretation.

6.3 Implications for Predictive Monitoring

The predictive monitoring approach used in this study shows that simple statistical methods can provide useful insight into system health. Rolling variance and threshold-based checks helped identify periods of increased variability, possible flat-line behaviour, and communication degradation.

These methods do not require complex machine-learning models and are suitable for early diagnostic monitoring. However, distinguishing between true environmental changes and sensor faults remains challenging. For example, increased temperature variance may reflect genuine environmental variation rather than sensor malfunction. Therefore, multiple indicators should be considered together when interpreting system reliability.

6.4 Digital Twin Perspective

The combination of real-time data streaming, live visualisation, mobile monitoring, and offline analysis forms an initial digital-twin-inspired representation of the indoor environment. The Grafana dashboard and iOS application provide real-time representations of sensor values, while the analytical pipeline reconstructs historical behaviour, identifies anomalies, and evaluates forecasting performance.

Although the current implementation is relatively simple, it demonstrates the core idea of a digital twin: integrating physical sensing with digital analytics to create a dynamic representation of a real-world system.

6.5 Forecasting Model Comparison

The forecasting results show that model performance depended on the variable being predicted. The seasonal-naive baseline performed best for temperature, suggesting that the previous daily pattern was already a strong predictor. In contrast, decomposition-based ARIMA produced lower errors for humidity and RSSI, indicating that modelling residual behaviour can be useful for some signals. However, the short 24-hour test period limits the strength of this comparison. A stronger evaluation would require longer data collection and rolling-origin validation.

6.6 Limitations

Several limitations should be considered when interpreting the results.

First, the duration of the dataset is relatively short, which restricts the ability to analyse long-term trends and seasonal effects beyond daily cycles. Second, the presence of inactive sensors reduces the dimensionality of the dataset and limits the scope of multivariate analysis.

Third, the system relies on a single sensing device, which may not fully capture spatial variability within the environment. Additionally, some anomalies may be caused by hardware or communication issues rather than actual environmental changes.
Finally, although forecasting models were explored, the analysis remains limited by the short dataset duration and the use of a single sensing device. More extensive data would be required to validate the predictive models more robustly.

6.7 Future Work

Future improvements could focus on both system design and analytical methods. Deploying multiple sensing nodes would allow spatial analysis and provide a more comprehensive representation of the environment. From a data science perspective, further optimisation of forecasting models could enhance predictive capability. Future work could extend the system towards a fully developed digital twin by incorporating predictive simulation and closed-loop control mechanisms. This would involve modelling the dynamic response of indoor environmental variables under different conditions and enabling automated actuation based on sensor inputs. Integrating such capabilities would allow the system not only to monitor and analyse the environment, but also to predict and optimise its behaviour in real time.

7. Conclusion

This project presented a smart home sensing system that combines IoT data collection, real-time visualisation, time-series analysis, and a simplified digital-twin framework. Sensor data generated by an ESP32-based smart farmhouse platform was transmitted through MQTT, stored, visualised, and analysed using Python workflows. The results showed clear daily patterns in light intensity and temperature. Lagged cross-correlation indicated that light changes were associated with later temperature variation, which is consistent with delayed thermal response. Temperature and humidity showed more synchronous behaviour. The project also identified important data-quality and reliability issues. Some sensor channels remained constant, event-based signals required native-resolution interpretation, and RSSI anomalies indicated temporary communication degradation. Rule-based monitoring and residual-based anomaly detection showed how simple statistical methods can support sensor-health assessment. Forecasting experiments using decomposition-based ARIMA showed mixed performance. The approach improved short-term prediction for humidity and RSSI, but not for temperature, where a seasonal-naive baseline performed better. Therefore, forecasting results were treated as exploratory.
Overall, the project demonstrates the feasibility of combining real-time IoT telemetry, time-series analytics, and predictive monitoring within a digital-twin-inspired framework for monitoring, interpretation, and early reliability assessment in smart environments. Future work should include longer data-collection periods, additional sensor nodes, improved calibration, and closed-loop control mechanisms to support the development of a more complete digital twin.

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